AETHERA
CASE 3: POLLUTANT TO PRODUCT

From Rice Husk Ash to Engineered Silicon

We earn the right to scale. RHA → silica → engineered silicon. Capital follows proof.

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Turning rice husk ash into battery-relevant silicon

Two pathways and three kill gates: we scale only as each gate passes.

The Problem

WASTE GENERATION
Rice-husk combustion leaves behind silica-rich ash that requires handling and disposal.

VALUE LOSS
Potentially recoverable silica remains trapped in a low-value waste stream.

TECHNOLOGY GAP
Extracting and purifying RHA into battery-grade silicon requires multiple process steps and quality controls.

ENVIRONMENTAL DILEMMA
Environmental Trade-off: Lower energy, chemical and carbon footprint.

Problem At Hand

CAN WASTE BECOME A RESOURCE, WITHOUT CREATING A NEW PROBLEM ?

OUR CORE SOLUTION

We turn the ash left after rice-husk burning into high-purity silica first, then into engineered silicon for battery anodes. Silica sales keep us earning while we prove the material, the customer and the economics. We invest further only as each gate passes. If it works, India's anode makers get a local alternative to imported silicon-based anode material.

Solutions Proposed



RECOVER
- Collect ash from existing rice-mill combustion
- Extract and purify silica (SiO_2)

UPGRADE
- Silica → Engineered Silicon
- Convert purified SiO_2 → Si / Si-C
- Optimise purity, structure and yield

QUALIFY
- Test electrochemical performance
- Validate customer specifications and economics

Two Pathways

Pathway 1 | PURIFIED SILICA

1. RHA → High-purity SiO_2
2. Lower technical complexity
3. Early commercial outlet
4. Establishes purification economics

Pathway 2 | ENGINEERED SILICON

1. SiO_2 → Si / Si-C
2. Higher value battery application
3. Material AND customer qualification
4. To be scaled only after 3 kill gates

Investment Logic

RHA
↓
PURIFIED SiO_2
↓
Si / Si-C
↓
BATTERY ANODE

LOWER RISK → HIGHER VALUE

Silica revenue → Battery-material upside

Three Kill Gates

Materials

Can we make a battery-relevant material?
- Purity & composition
- Structure & yield
- Anode performance

PASS → Material meets required specification

Customer

Can we make it competitively?
- Cost per kg
- Input & energy costs
- Price of material displaced

PASS → Cost ≤ value of displaced material

Economics and Carbon

Does it deliver real environmental benefit?
- Process energy
- Magnesium & chemical inputs
- Grid-related emissions

PASS → Net CO_2e benefit remains positive

The feedstock already exists, in boilers that burn rice husk every day

RHA comes from mills that already burn husk, so we add no new burning. We start with one pilot cluster and two backups.

01 What is RHA?

Rice Husk Ash (RHA) is the ash left after rice husk is **burned in mill boilers.**

- **Mostly silich (typically 90%+), with amorphous form.**
- **Reactive and suitable as a silica-rich feedstock.**
- **~4.8 Mt/year in India**' (*To be verified)*

02 Why RHA?

1	**Fly ash**	›	Higher Al/Fe content
2	**Bagasse ash**	›	Already refined silica
3	**Silica fume**	›	Already refined silica
4	**PV kerf waste**	›	Already silicon, different industry
5	RHA offers a high-silica waste stream directly available from rice mills.		

03 How we capture it

Collected from the boiler ash discharge at mills that already burn husk.

No additional burning is required.

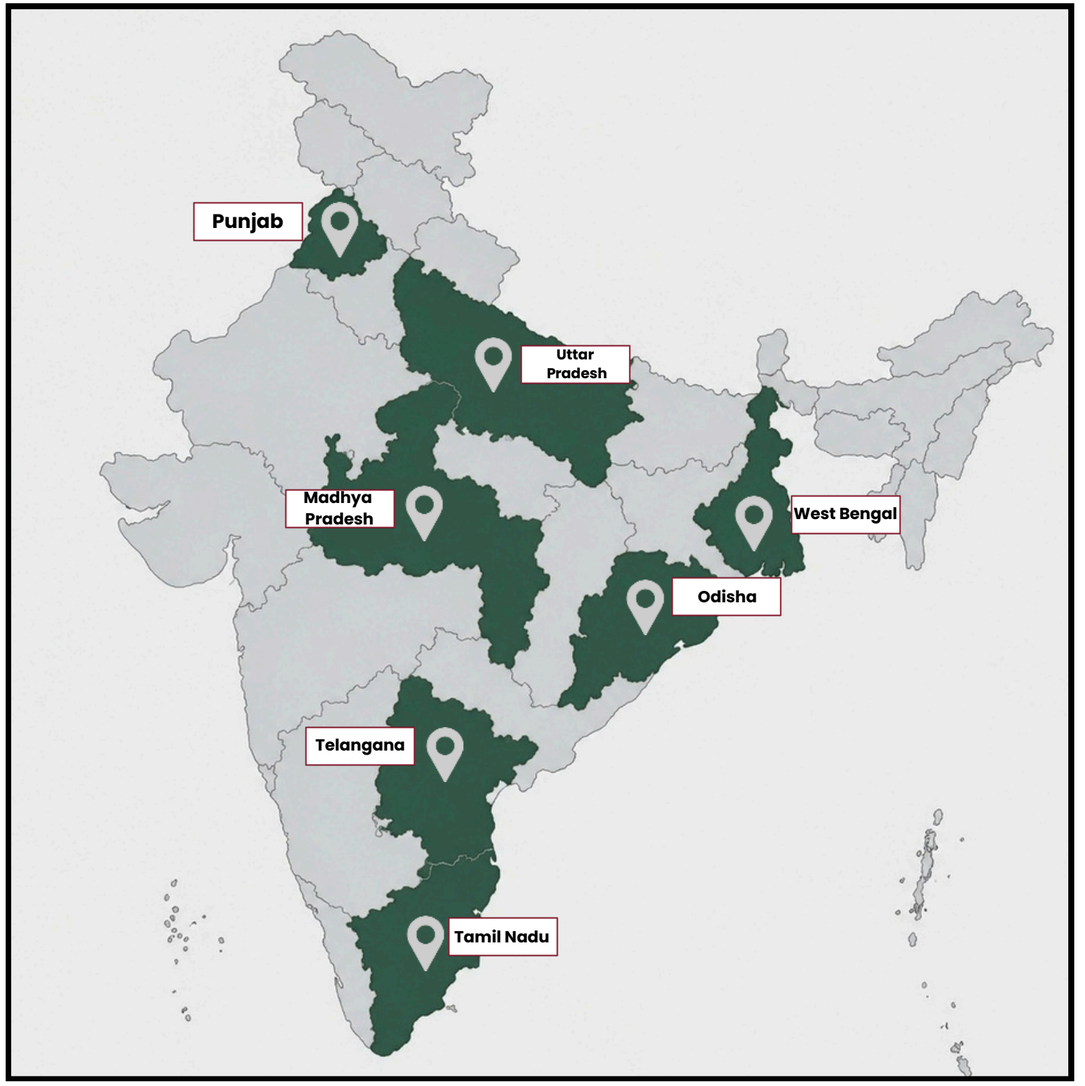
04 Cluster Division

Score candidate rice-milling clusters on four criteria:

- Mill density
- Boiler type & ash quality
- Distance to plant
- Willingness to sign supply contracts

Output:
1 pilot cluster + 2 backup clusters

THE INDIA SOURCING LANDSCAPE



1. UTTAR PRADESH — 20.76 MT
2. TELANGANA — 17.45 MT
3. WEST BENGAL — 16.02 MT
4. PUNJAB — 14.36 MT
5. CHHATTISGARH — 10.37 MT
6. ODISHA — 9.53 MT
7. MADHYA PRADESH — 9.11 MT
8. BIHAR — 8.30 MT
9. ANDHRA PRADESH — 8.16 MT
10. TAMIL NADU — 7.09 MT

Four steps turn rice husk ash into battery-relevant silicon

Each step gives us something we can test, and silica can be sold before silicon is ready.

This end-to-end sustainable value chain transforms local agricultural waste into high-performance, battery-grade silicon powder for India's growing energy storage ecosystem



RICE MILL

1. Sourcing agricultural waste husks locally from rice milling hubs at minimal cost.
2. Securing a decentralized, continuous supply of raw biomass feedstock.

BOILER

1. Controlled combustion of husks in biomass boilers to extract amorphous silica.
2. Generating raw Rice Husk Ash (RHA) rich in silica under optimized temperatures.

SILICA RICE ASH

1. Initial collection of carbon-rich ash byproduct from industrial biomass boilers.
2. Raw material staging containing concentrated amorphous silica particles.

PURIFY

1. Chemical acid leaching (HCl wash) to dissolve and remove metallic impurities.
2. Extracting high-grade silica (SiO2) by stripping away iron, potassium, and sodium.

SIO_2

1. High-purity silica intermediate ready for thermal reduction furnaces.
2. Chemically cleansed oxide stage meeting precise precursor specifications.

REDUCE

1. High-temperature furnace processing using a clean reductant to strip oxygen.
2. Converting purified silica into raw elemental silicon metal.

REFINED SILICON

1. Raw silicon output carrying baseline purity standards before final resizing.
2. Intermediate metallic silicon product ready for specialized down-scaling.

ENGINEER

1. Advanced nano-milling, particle sizing, and surface classification.
2. Achieving battery-grade sub-micron specs for composite blending.

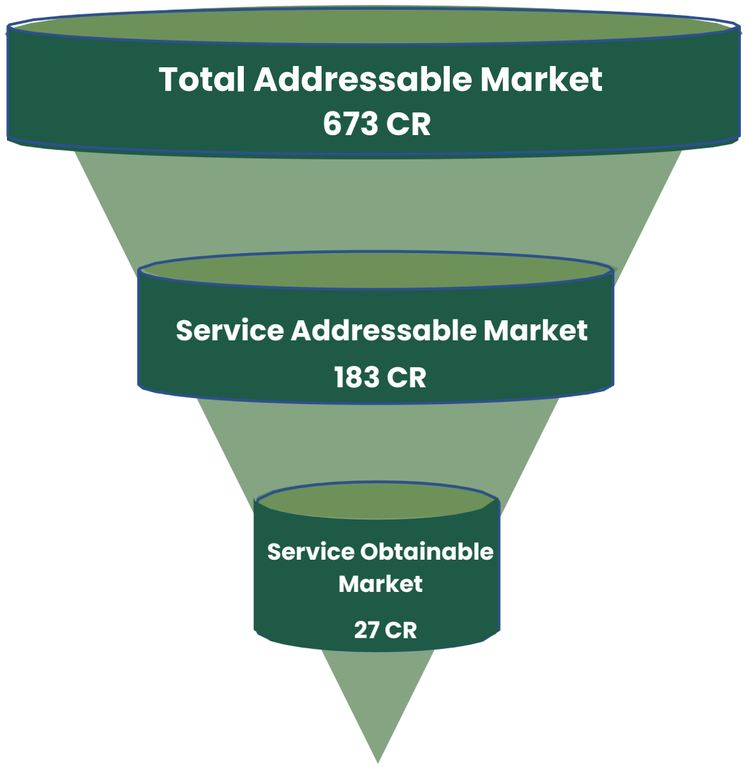
ANODE MAKER

1. Delivering engineered powder directly to lithium-ion anode-material compounders.
2. Establishing local B2B supply channels for high-performance EV battery anodes.

A small, winnable slice of India's battery-anode market

Indian anode makers want local, secure supply, and our feedstock is waste.

Devising an Obtainable Market Size for high purity engineered silicon powder for battery anode material



Refer to Annexure

Analysing Market for Purified Silica

Targeting domestic anode compounders through sustainable waste valorisation establishes our clear competitive wedge in India's battery ecosystem

Segmentation

- **By Application:** Separating solar polysilicon, rubber filler silica, and energy storage lithium-ion battery anodes.
- **By Supply Chain:** Distinguishing upstream raw ash suppliers, mid stream anode material compounders, and downstream cell giga factories

Targeting

Primary Target: Domestic Indian **Lithium-ion Anode-Material Manufacturers** because its a **high growth segment** actively seeking local, cost-effective, and **secure silicon-carbon composite** feedstock to replace expensive imports.

Positioning

For **Indian anode** manufacturers, our engineered silicon powder transforms agricultural rice husk ash into a **high performance**, **sustainable material**. We deliver **battery grade purity** with a **similar cost** and **carbon footprint of imported silicon**, securing a local **"Make in India"** supply chain for advanced batteries.

Abundant low-cost waste feedstock and technical barriers offset high buyer power, creating a high-margin wedge for local early movers.

	LOW	HIGH
THREAT OF NEW ENTRY		
COMPETITIVE RIVALRY		
BARGAINING POWER OF SUPPLIERS		
BARGAINING POWER OF BUYERS		
THREAT OF SUBSTITUTION		

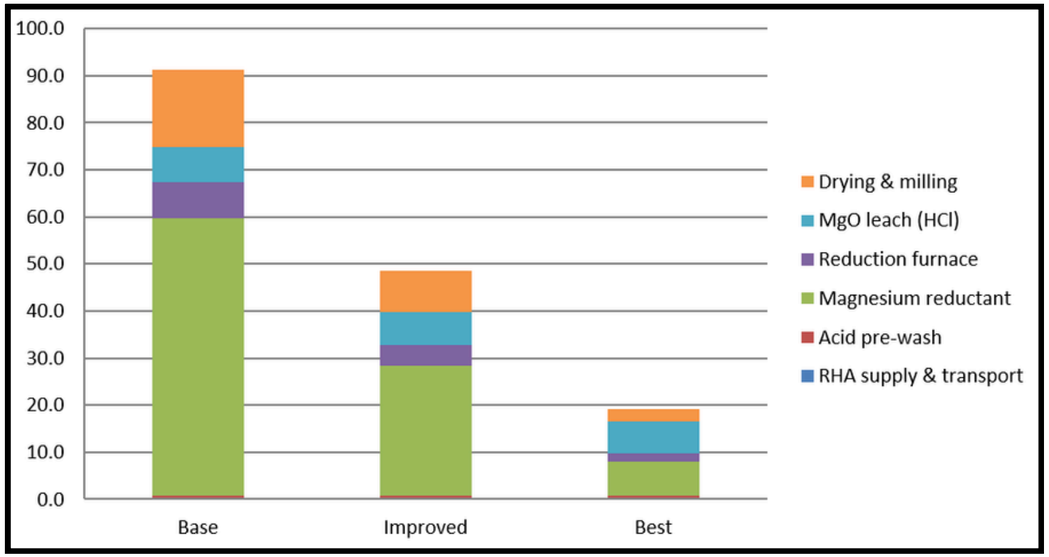
The above conclusion show that leveraging zero-cost agricultural waste (RHA) provides a structural cost advantage and first-mover defense against high-priced global substitutes in India's growing battery ecosystem.

Most of the carbon is in the magnesium, so that's where we cut it

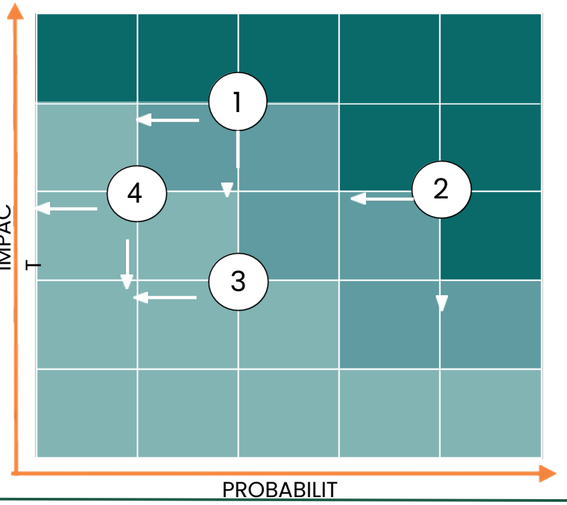
Renewable power and recycled magnesium cut the footprint by about 80% from Base to Best.

Quantifying our carbon footprint reduction from magnesium optimisation and renewable energy adoption

Footprint at various stages	Base	Improved	Best
RHA supply& transport	0.0	0.0	0.0
Acid pre-wash	0.8	0.7	0.7
Magnesium Reductant	58.9	27.6	7.3
Reduction Furnace	7.7	4.4	1.7
MgO leach(HCL)	7.3	7.1	6.8
Drying and Milling	16.6	8.7	2.5



Most of the carbon in turning **rice husk ash** into **high-purity silicon** comes from **magnesium and energy-intensive processing**, so targeted action can cut it sharply. Base uses grid power and fresh magnesium; Improved adds **50% renewable power** and **partial magnesium recycling**; Best uses **90% renewable power** and **80% recycling**. Moving from **Base to Best** makes the route a **competitive low-carbon**, **circular source of silicon** for **India's battery anode manufacturers**.



Proactive mitigation of environmental, supply, and operational risks ensures robust project viability and regulatory compliance.

	Risk	Why it matters?	Mitigation
1	Magnesium reductant carbon	65% of Base footprint; decides whether the route beats MG-Si	Buy low-carbon Mg with an EPD; build $MgCl_2$ electrolysis recycling (target 50% → 80%); contract Mg supply early
2	Grid power in furnace and milling	32.4 kWh/kg at 0.727 kg/kWh; milling alone is 18%	Green PPA (50% → 90%); lower-energy milling and classification; heat recovery in furnace
3	HCl effluent and MgO/$MgCl_2$ salts	About 8 kg HCl and 3.4 kg MgO per kg Si; acidic, saline wastewater	Closed-loop acid and water recovery; neutralisation and ETP; sell MgO as co-product; comply with CPCB/SPCB consent conditions
4	RHA quality and supply variability	90% SiO_2 is assumed; real ash varies with boiler temperature, carbon content and metals, which affects purity	Long-term contracts with biomass boilers; incoming QC (SiO_2, carbon, K, Na, Fe); acid pre-wash; blend lots

Click here to view detailed model of reduction in carbon footrprint:

The economics work, and we can stop at every gate

Base case, with a Low to High range. Pathway 2 goes ahead only if every gate passes.

Click here to view detailed economic model

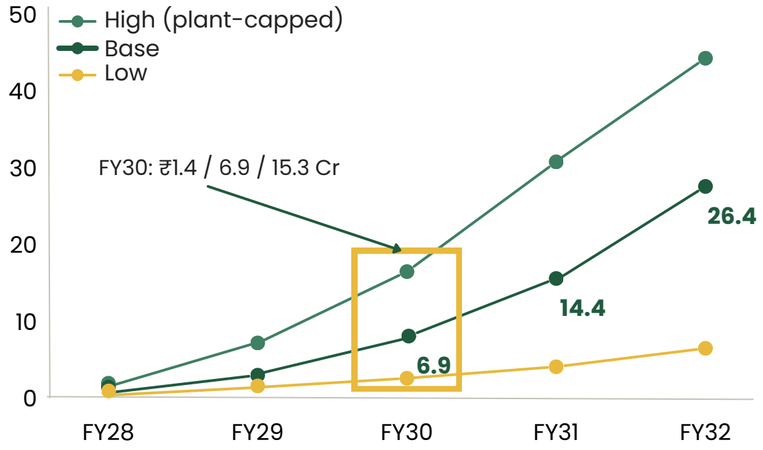
RINOLYST research and innovation catalyst

Base case · ₹ Crore · USD 30/kg at ₹88/$

Modelled Economics

Revenue grows in all three cases, reaching ₹5 to ₹43 Cr by FY32

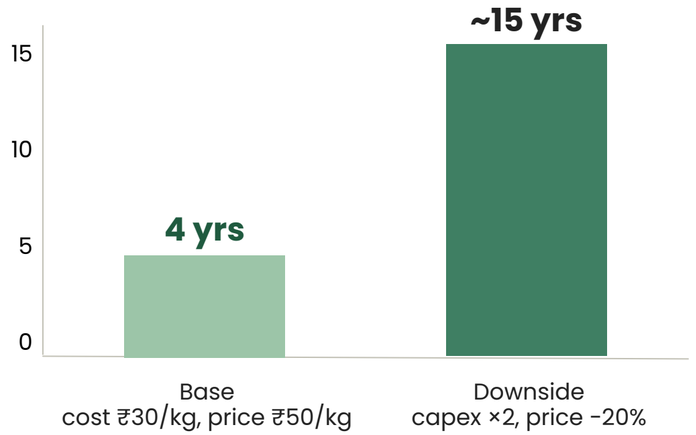
Obtainable revenue, ₹ Cr (Low / Base / High)



We earn revenue even in the Low case, so the plan doesn't need the High case to work.

Pathway 1 pays back in about 4 years, or about 15 in the worst case

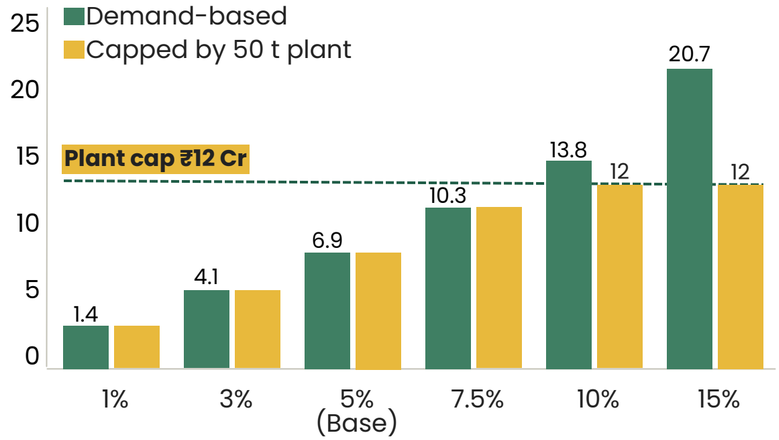
Years to recover capex



Selling silica alone gets the capex back in about 4 years in the Base case.

If adoption beats the plan, our 50 t plant limits FY30 revenue to ₹12 Cr

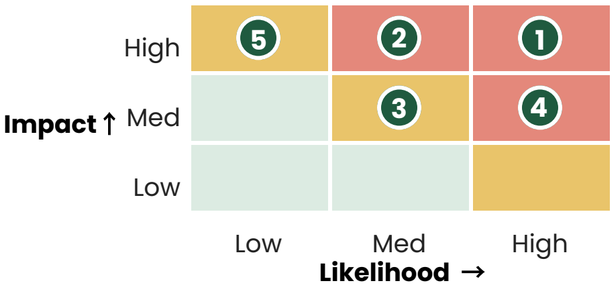
FY30 revenue vs Si-anode adoption, ₹ Cr



Past ~10% adoption, the plant becomes the limit. We add capacity only once customers have qualified us.

Safeguards

Risk heat map



Risk and response

1. **Slow customer qualification: silica sales** keep us earning, and we add no capacity before the Customer Gate.
2. **Selling price lower than assumed:** even at **USD 22/kg** we still earn, and the Economics & Carbon Gate checks ₹/kg before we scale.
3. **Magnesium costs more or yield falls short:** we work out the **break-even price** at the Economics & Carbon Gate.
4. **Competition from Cellark and Chinese makers:** we **supply anode makers locally** and stay out of finished anodes.
5. **Dust and acid handling:** a **closed-loop process** and a **controlled burn temperature** (to be confirmed with an engineer).

Gates: the stop-loss

Gate 1: Material
Purity, structure, yield and battery performance hit the target

Gate 2: Customer
Anode makers approve our kg-scale samples

Gate 3: Economics & Carbon
Cheaper per kg than the material it replaces, with a net CO_2e saving

We buy only ash from husk that mills already burn. We check boiler fuel records, fund no new husk burning, and cap purchases at past output.

We earn the right to scale

RHA → silica → engineered silicon. Capital follows proof.

From RHA to Commercial Sale

Two pathways convert RHA into value: purified silica provides near-term revenue, while engineered silicon advances only as technical, commercial and environmental risks are resolved.

Pathway 1: PURIFIED SILICA — RHA → Purification → Silica qualification → Buyer → Revenue — REVENUE BRIDGE

Pathway 2: ENGINEERED SILICON — Silica → Conversion → Performance → Buyer trials → Scale

- **Materials** — Purity • yield • performance
- **Customer** — kg-scale • anode qualification
- **Economics & Carbon** — ₹/kg • net CO_2e

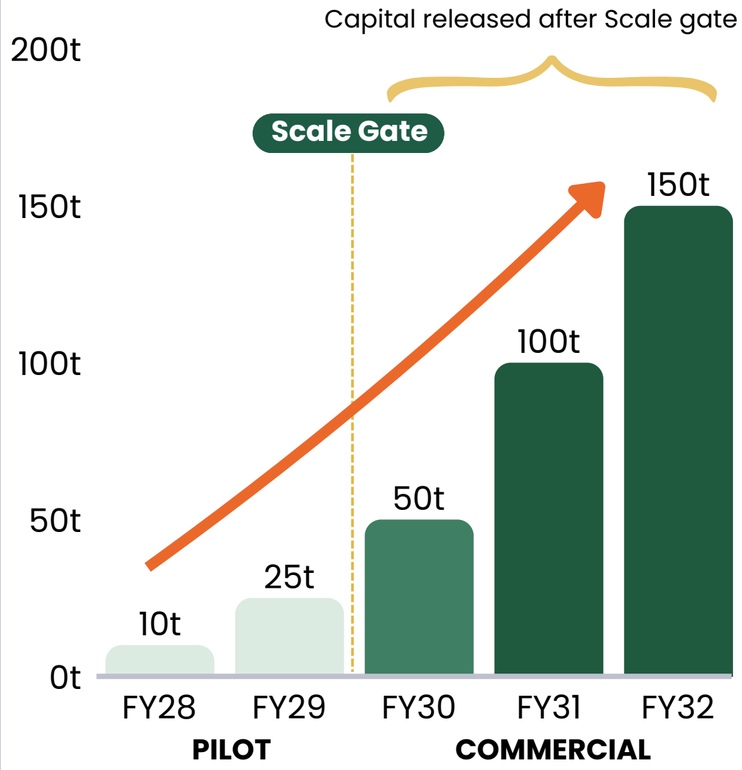
Feasibility Sequencing

TECHNICAL PROOF			CUSTOMER PROOF	COMMERCIAL SCALE	
1	2	3 Gate 1: Material	4 Gate 2: Customer	5 Gate 3: Economics & Carbon	6
Feedstock	**Purify**	**Convert**	**Customer Qualify**	**Scale Gate**	**Scale**
Secure RHA supply	Silica purity	Engineered Si	Anode-maker trials	₹/kg economics	Commercial plant
+ contamination baseline	+ recovery yield	+ performance validation	+ kg-scale samples	+ net CO_2e benefit	+ replicate clusters

Each gate unlocks the next level of capital commitment

Modelled Capacity

Si powder, tonnes per year



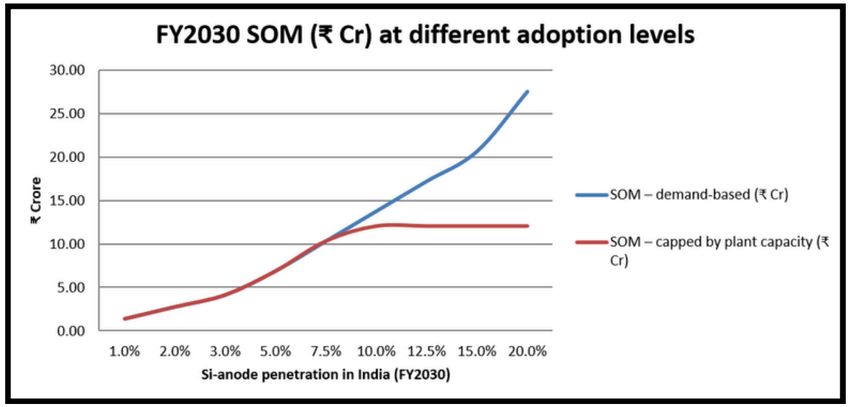
FY2030 MARKET SIGNAL

28.6 t	₹6.9 Cr	143 t
SOM Si powder	SOM value	RHA required

APPENDIX

TAM SAM SOM MODEL

Paticulars	FY2027	FY2028	FY2029	FY2030	FY2031	FY2032
India Li-ion battery demand	54	71.8	95.5	127	146.1	168
Anode active material demand	59,400	78,993	105,049	139,700	160,655	184,753
Si-bearing anode penetration (scenario-adjusted)	1.00%	2.00%	3.50%	5.00%	6.50%	8.00%
Si-bearing anode volume	594	1,580	3,677	6,985	10,443	14,780
TAM volume – silicon powder demand	59.4	158	367.7	698.5	1,044.30	1,478.00
Selling price	2,640	2,561	2,484	2,409	2,337	2,267
TAM (value)	15.7	40.5	91.3	168.3	244.1	335.1
SAM filters						
× India-made cell share	30%	40%	55%	65%	70%	75%
× Technology fit	60%	60%	60%	60%	60%	60%
× Open to new / domestic supplier	40%	50%	60%	70%	70%	70%
SAM volume	4.3	19	72.8	190.7	307	465.6
SAM (value)	1.1	4.9	18.1	45.9	71.8	105.5
SOM build-up						
Obtainable share of SAM (scenario-adjusted)	0.00%	5.00%	10.00%	15.00%	20.00%	25.00%
Demand-based obtainable volume	0	0.9	7.3	28.6	61.4	116.4
Plant capacity	0	10	25	50	100	150
SOM volume (lower of demand-based and capacity)	0	0.9	7.3	28.6	61.4	116.4
SOM (value)	0	0.24	1.81	6.89	14.35	26.39
Ratios						
SAM as % of TAM	7.20%	12.00%	19.80%	27.30%	29.40%	31.50%
SOM as % of SAM	0.00%	5.00%	10.00%	15.00%	20.00%	25.00%
SOM as % of TAM	0.00%	0.60%	1.98%	4.10%	5.88%	7.88%
Feedstock cross-check						
RHA feedstock required for SOM volume	0	5	36	143	307	582



Key Assumptions	
USD/INR exchange rate (₹)	88
FY2027 India Li-ion demand (GWh)	54
Demand growth after FY2030 (% p.a.)	15.00%
Implied FY2027–FY2030 demand CAGR	33.00%
Anode active material per GWh (tonnes/GWh)	1,100
Silicon content in Si-bearing anodes (% by weight)	10.00%
Selling price FY2027 (₹/kg)	2,640
Annual price change (% p.a.)	-3.00%
RHA required per tonne of Si powder (tonnes)	5
Rupees per Crore	10,000,000

ANNEXURE

RBI, Handbook of Statistics on Indian States 2024-25, Table 58: State-wise Production of Foodgrains, Rice (data from the Ministry of Agriculture & Farmers' Welfare)

CareEdge Ratings, Lithium-ion Battery Storage report, Nov 2024.

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ULLAL, SHIVARAMU & AVEEN (2022), "RICE HUSK ASH UTILIZATION, COMPOSITION AND PROPERTIES: A BRIEF REVIEW", J. MET. MATER. MINER. 32(4), 37–46. DOI 10.55713/JMMM.V32I4.1544

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TAM, SAM, SOM Model

International Magnesium Association LCA study, summarised in Light Metal Age

ICEA–Accenture report on India's Li-ion market (as reported by ANI/Tribune)

CEEW Centre for Energy Finance (as reported by pv magazine India).

THANK YOU